

Optional Proofs — Progress Update

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Weak-Subjectivity Checkpoint Execution Proof Sync

When joining the network a node must validate execution payloads from weak-subjectivity checkpoint to head:

Sync policy	Proof bytes over one WS window
1 proof per payload	~26.9 GiB
2 proofs per payload	~53.9 GiB
4 proofs per payload	~107.8 GiB

Assuming the Electra WS reference period of 3,532 epochs, 32 slots per epoch, 113,024 slots, and 250 KiB per execution proof.

- **Naive sync scales linearly** with both the weak-subjectivity window length and the proof policy.
- **Recursive direction** — add a `RecursiveExecutionProof` layer above base `ExecutionProof`.
- **Chain-prefix authentication** — use a Merkle tree or MMR over proven beacon roots to authenticate that a block is part of the proven chain prefix.
- **Network simplification** — a single proof for the chain tip can authenticate the full prefix, including the checkpoint-to-tip range, removing most requirements for historical execution-proof req/resp.
- **Next steps** — spec the chain-prefix / checkpoint-membership scheme, then implement the proof-sync path.